Short questions

|!|BCESC0160000001|!|

Complete the following paragraph using appropriate words. (4 marks)

Sense organs are groups of (a)_____ cells. They response to specific (b)_____ and relay information as electrical signals called (c)_____ to the (d)_____ for interpretation to produce sensations.

##

(a) receptor / sensory (1)
(b) stimuli (1)
(c) nerve impulses (1)
(d) brain (1)

(4 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1

|!|BCESC0160000002|!|

Complete the following table about human senses. (3 marks)

	Sense	Sense organ	Stimulus
(a)	sight		
(b)		ear	
(c)			chemical

##

(a) eye, light (1)
(b) hearing, sound (1)
(c) taste, tongue
OR
smell, nose (1)

(3 marks)

© 2019 Aristo Educational Press Ltd.

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1

|!|BCESC0160000003|!|

Complete the following paragraphs using appropriate words. (5 marks)

The eyeball is situated in a bony cavity in the (a)_____. The wall of the eyeball is attached to three pairs of (b)_____ which contract and relax to allow the eyeball to rotate.

Light rays entering the eye are refracted by the (c)_____ and the lens. This brings the light rays into focus on the (d)_____. The shape of the lens is altered by the (e)_____.

##

(a) skull (1)
(b) eye muscles (1)
(c) cornea (1)
(d) retina (1)
(e) ciliary body / ciliary muscles (1)

(5 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1

|!|BCESC0160000004|!|

Complete the following paragraph using appropriate words. (5 marks)

When looking at near objects, the circular ciliary muscles (a)_____. The perimeter of the ciliary body becomes (b)_____. This (c)_____ the tension in the suspensory ligaments. The lens becomes (d)_____ due to its elasticity and bends light rays more, so that the image of the near object can be focused sharply onto the (e)_____.

© 2019 Aristo Educational Press Ltd.

##

(a) contract (1)

(b) smaller (1)

(c) reduces (1)

(d) thicker / more convex (1)

(e) retina (1)

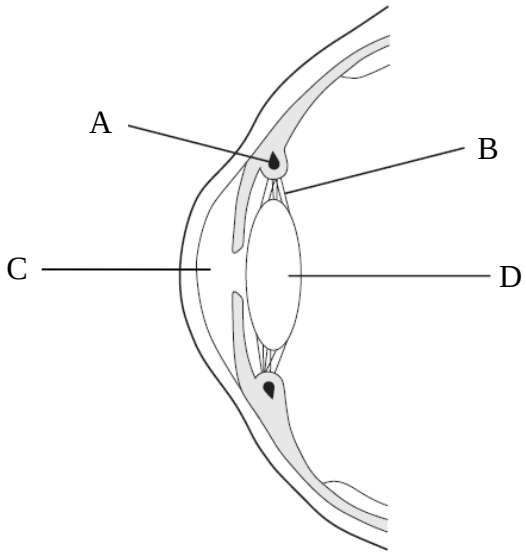
(5 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1

|!|BCESC0160000005|!|

The diagram below shows a section of the front part of a human eye.



(a) Name A and B. (2 marks)

(b) Give ***one*** function of the liquid in chamber C. (1 mark)

© 2019 Aristo Educational Press Ltd.

(c) State and explain ***two*** adaptations of structure D. (4 marks)

##

(a) A: ciliary muscles / ciliary body (1)

B: suspensory ligament (1)

(b) Any ***one*** of the following: (1)

- supplies oxygen and nutrients to the conjunctiva, the cornea and the lens
- maintains the shape of the eyeball
- refracts light onto the retina

(Or other reasonable answers)

(c) Any ***two*** of the following: (4)

- It is transparent / contains no blood vessels (1)
  and so light can pass through without obstruction. (1)
- It is biconcave in shape (1)
  and can refracts light onto the retina. (1)
- It is elastic (1)
  and so it can change its curvature / thickness / shape to focus light from different distances. (1)

(Or other reasonable answers)

(7 marks)

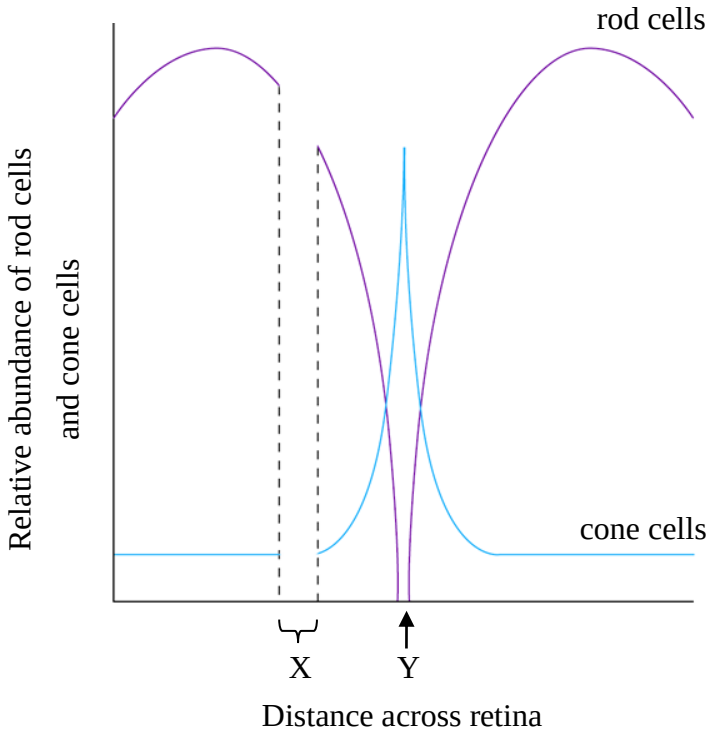
##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	4

© 2019 Aristo Educational Press Ltd.

!|BCESC0160000006|!|

The diagram below shows the distribution of rod cells and cone cells in the retina.



(a) Which part of the retina dose X represent? Explain your answer. (2 marks)

(b) Which part of the retina dose Y represent? Explain your answer. (2 marks)

(c) What are the functions of rod cells and cone cells? (2 marks)

##

(a) Blind spot (1)

There are no rod cells or cone cells / photoreceptors / light-sensitive cells. (1)

(b) Yellow spot (1)

It contains the highest density of cone cells but no rod cells. (1)

© 2019 Aristo Educational Press Ltd.

(c) Rod cells are responsible for black and white vision / vision in dim light. (1)
Cone cells are responsible for colour vision. (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Analysing data	2
(c)	Understanding basic concepts	2

|!|BCESA0160000007|!| ★★

(a) Where is the blind spot located in the eye? (1 mark)

(b) A student investigated the blind spot in his right eye. He closed his left eye and look at Figure 1 printed on a piece of paper.

Figure 1



He moved the piece of paper towards him until the dot seemed to disappear. Figure 2 shows what he saw.

Figure 2



(i) Explain this phenomenon. (3 marks)

© 2019 Aristo Educational Press Ltd.

(ii) If the student had kept his left eye open, the dot would not have disappeared. Explain why. (2 marks)

_____

_____

_____

##

(a) The blind spot is located in the area of the retina where the optical nerve leaves the eye. (1)

(b) (i) The image of the dot fell on the blind spot (1)
where there are no photoreceptors. (1)
No nerve impulses could be generated to stimulate the visual centre of the brain. (1)
Therefore, he could not see the image of the dot.

(ii) The two eyes see from slightly different angles. (1)
The image of the dot would not fall on the blind spot on both eyes at the same time. (1)

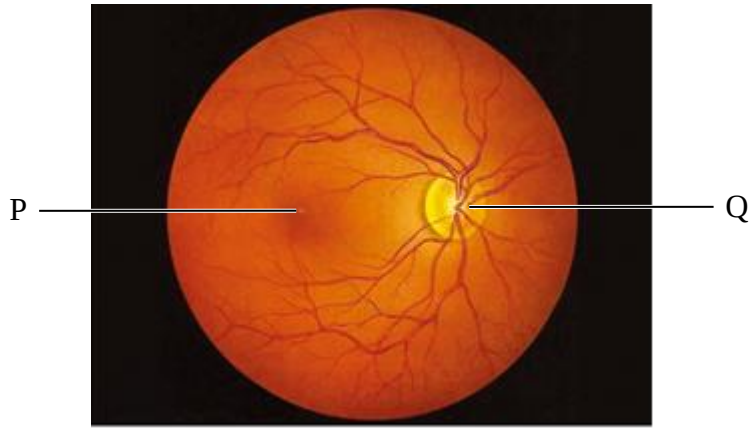
(6 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)(i)	Applying concepts	3
(b)(ii)	Critical thinking	2

|!|BCESB0160000008|!| ★

The photograph below shows the retina of a human eye.



© 2019 Aristo Educational Press Ltd.

(a) Identify P and Q. (2 marks)

(b) Explain the difference in the vision produced when the image of a small object is formed on P and Q. (4 marks)

##

(a) P: yellow spot (1)
Q: blind spot (1)

(b) The image formed on P can be seen most clearly (1)
because P contains the highest density of cone cells. (1)
The image formed on Q cannot be seen (1)
because there are no photoreceptors at Q. (1)

(6 marks)

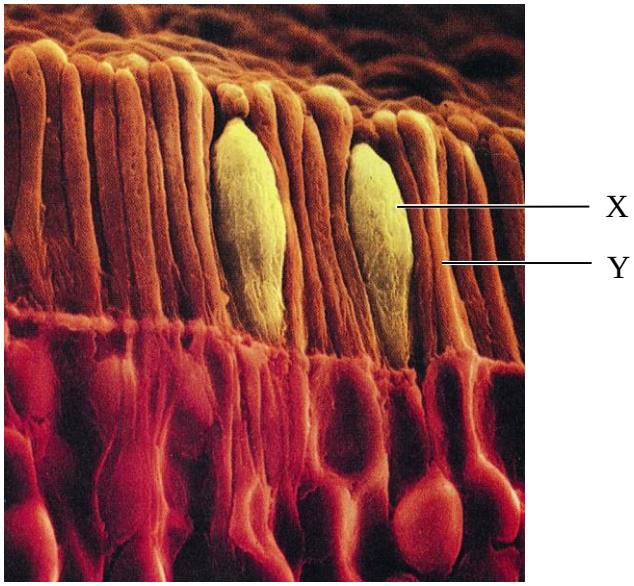
##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Understanding basic concepts	4

© 2019 Aristo Educational Press Ltd.

|!|BCESB0160000009|!| ★

The photomicrograph below shows part of the human retina.



(a) Identify cells X and Y. (2 marks)

(b) In very dim light, a person cannot see the colour of an object as clearly as in bright light. Explain this with respect to the functioning of cells X and Y. (4 marks)

##

(a) X: cone cell (1)
Y: rod cell (1)

(b) Cell X is responsible for colour vision (1)
while cell Y cannot distinguish colours. (1)
Cell X is only stimulated by strong light, and (1)
a smaller number of cell X is stimulated in dim light than in bright light. / a larger number of cell X is stimulated in bright light than in dim light. (1)

(6 marks)

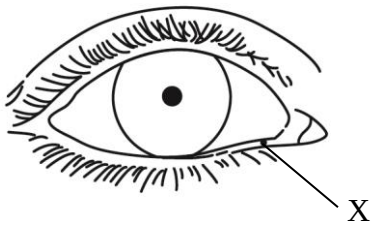
##

© 2019 Aristo Educational Press Ltd.

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Applying concepts	4

|!|BCESC0160000010|!|

The diagram below shows the front view of a human eye.



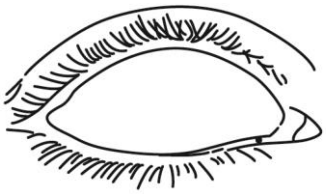
(a) State ***one*** function of the liquid secreted from structure X. (1 mark)

_____

_____

(b) The eye experienced a decrease in light intensity.

(i) Complete the drawing below to show how the eye would response. (1 mark)



(ii) Explain how this response is brought about. (2 marks)

_____

_____

_____

© 2019 Aristo Educational Press Ltd.

(iii) Explain the importance of this response. (2 marks)

_____

_____

_____

##

(a) It keeps the eye surface moist and clean. / It helps reduce friction when during blinking. / It contains lysozymes which can kill bacteria. (1)

(b) (i)



(ii) The circular muscles of the iris relaxed (1)
while the radial muscles contracted. (1)
Thus, the pupil dilated.

(iii) This increased the amount light entering the eye (1)
to increase stimulation of the photoreceptors for a clearer vision in low light intensities. (1)

(6 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)(i)	Presenting information	1
(b)(ii)	Understanding basic concepts	2
(b)(iii)	Understanding basic concepts	2

© 2019 Aristo Educational Press Ltd.

|!|BCESA0160000011|!| ★★ **STSE**

Photograph I shows the appearance of a person's eye in dim light. Photograph II shows the same eye just after a camera flash.



Photograph I



Photograph II

(a) What is the significance of this response of the eye? (2 marks)

(b) When we take photographs with the flash on, 'red eyes' may appear in the photographs (i.e. the pupils of the eyes become red). Explain this phenomenon. (2 marks)

(c) In cameras with a 'red-eye reduction' function, the flash goes off twice—once before the photograph is taken and then again to take the photograph. Explain how this works in preventing 'red eyes' in photographs. (2 marks)

##

(a) This limits the amount of light entering the eye (1)
so as to prevent over-stimulation / damage of the photoreceptors / retina. (1)

(b) Red light is reflected by the blood in the capillaries in the choroid (1)
and goes out through the pupil. (1)

(c) The first flash causes the pupil to constrict. (1)
This reduces the amount of light reflected off the choroid after the second flash. (1)

(6 marks)

© 2019 Aristo Educational Press Ltd.

##

	Skills assessed	Marks
(a)	Understanding basic concepts	2
(b)	Applying concepts	2
(c)	Applying concepts	2

|!|BCESB0160000012|!| ★ **STSE**

(a) Describe the changes which take place in the eye when a person walks out from a dim movie theater into the bright sunlight. (3 marks)

(b) Sunglasses are worn for protection as well as fashion. However, not all sunglasses can cut off ultra-violet light. Suggest why wearing such sunglasses in bright sunlight would do more harm to the eye than not wearing any sunglasses at all. (3 marks)

##

(a) The increase in light intensity cause the circular muscles of the iris to contract (1)
and the radial muscles to relax. (1)
The pupil becomes smaller / constricts. (1)

(b) Sunglasses reduce the intensity of the light entering the eye (1)
so the pupil will not constrict / constrict to smaller extent in bright sunlight. (1)
As a result, more UV light can enter the eye and cause damage. (1)

(6 marks)

© 2019 Aristo Educational Press Ltd.

##

	Skills assessed	Marks
(a)	Understanding basic concepts	3
(b)	Applying concepts	3

|!|BCESC0160000013|!|

John was reading the newspaper in the living room. Suddenly, he heard a loud sound and looked out the window to see what happened.

(a) Complete the following flow chart to show the major steps involved in hearing the loud sound. (4 marks)

sound waves being directed through the outer ear by the (i) _____

↓

vibration of the eardrum

↓

transmission through ear bones to reach the (ii) _____

↓

detection of the (iii) _____ movement by sensory hair cells

↓

transmission of nerve impulses through the (iv) _____ to the brain for interpretation

(b) Describe the changes that occurred in the eye when John looked up from the newspaper to look out the window. (3 marks)

_____

_____

_____

_____

##

(a) (i) auditory canal (1)
(ii) oval window / cochlea (1)
(iii) liquid / endolymph (1)
(iv) auditory nerve (1)

© 2019 Aristo Educational Press Ltd.

(b) The circular ciliary muscles relax. (1)
This increases the tension in the suspensory ligaments. / The suspensory ligaments become tighter. (1)
As a result, the lens is pulled to a thinner (less convex) shape to focus on distant objects. (1)

(7 marks)

##

	Skills assessed	Marks
(a)(i)	Understanding basic concepts	1
(a)(ii)	Understanding basic concepts	1
(a)(iii)	Understanding basic concepts	1
(a)(iv)	Understanding basic concepts	1
(b)	Understanding basic concepts	3

|!|BCESC0160000014|!|

A man suffers from an eye defect that he cannot see nearby objects clearly.

(a) Draw a ray diagram to show the path of light rays entering his eye under such a condition. (3 marks)

(b) What type of lens can used to correct this eye defect? (1 mark)

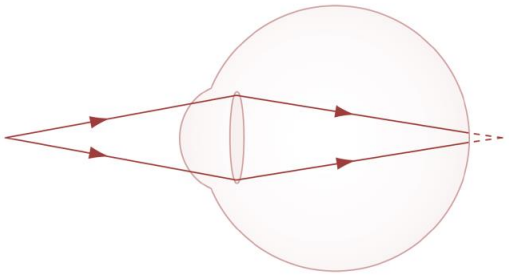
© 2019 Aristo Educational Press Ltd.

##

(a) Correct outline of the eyeball and lens (1)

Light rays from a nearby object diverging from a single point (1)

Light rays intersect behind the retain; dotted lines used to represent light rays behind the retina (1)



(b) Convex lens (1)

(4 marks)

##

	Skills assessed	Marks
(a)	Presenting information	3
(b)	Understanding basic concepts	1

|!|BCESC0160000015|!|

(a) State ***one*** cause of short sight. (1 mark)

(b) Draw a ray diagram to show how long sight can be corrected using appropriate lenses. (3 marks)

© 2019 Aristo Educational Press Ltd.

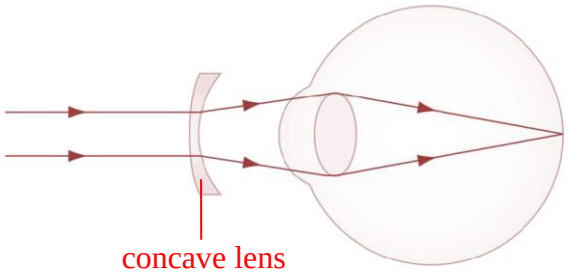
##

(a) The eyeball is too long. / The lens is too thick. (1)

(b) Concave lens correctly labelled (1)

Parallel light rays diverged by the concave lens (1)

Light rays intersect on the retina (1)



(4 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Presenting information	3

|!|BCESC0160000016|!|

For each of the ear structures listed in ***Column 1***, select from ***Column 2*** one phrase that matches it. Put the appropriate letter in the space provided. (3 marks)

Column 1		***Column 2***
eardrum	(a) _____	A. detects the direction of head movement
semicircular canal	(b) _____	B. transmits vibrations from the ear bones to the inner ear
oval window	(c) _____	C. releases the pressure from the cochlea into the middle ear
		D. converts sound waves into mechanical vibrations
		E. contains sensory hair cells that detect vibrations

© 2019 Aristo Educational Press Ltd.

##

(a) D (1)

(b) A (1)

(c) B (1)

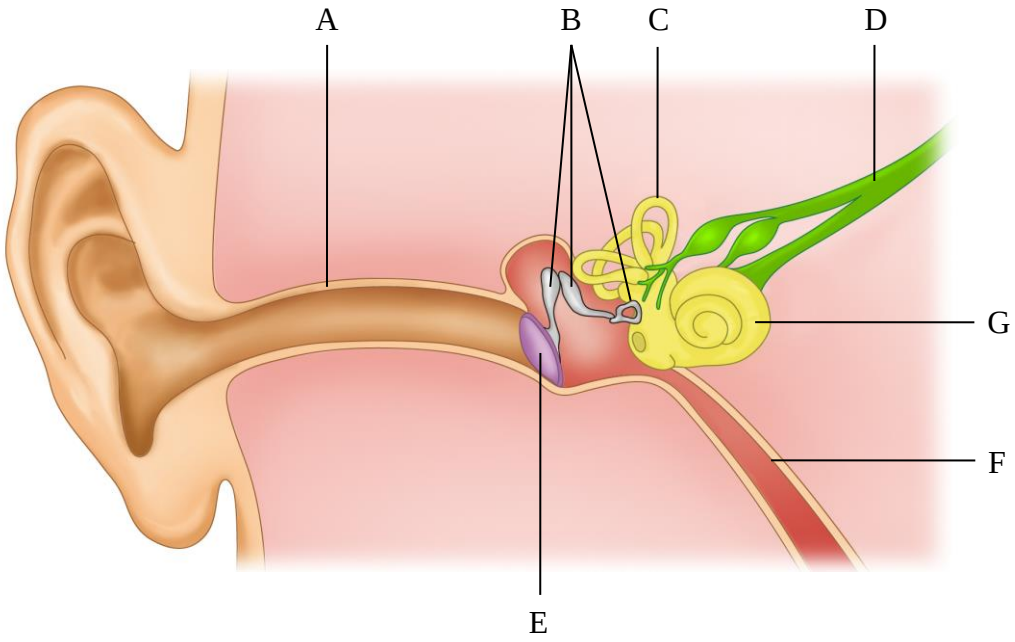
(3 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1

|!|BCESC0160000017|!|

The diagram below shows the human ear and its associated structures.



Use the letters in the diagram to complete the table below. (5 marks)

Function	Structure
(a) To detect head movements	
(b) To generate nerve impulses to the brain for hearing	
(c) To convert sound waves to vibrations	
(d) To transmit and amplify vibrations	
(e) To direct sound waves into the ear	

© 2019 Aristo Educational Press Ltd.

##

(a) C (1)

(b) G (1)

(c) E (1)

(d) B (1)

(e) A (1)

(5 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	1
(c)	Understanding basic concepts	1
(d)	Understanding basic concepts	1
(e)	Understanding basic concepts	1

|!|BCESC0160000018|!|

(a) How does a plant shoot respond to unilateral light? (1 mark)

(b) Explain this response of the shoot on a hormonal basis. (3 marks)

(c) Explain the significance of this response to the plant. (1 mark)

##

(a) The shoot bends towards the light. (1)

(b) Unilateral light causes auxins to be transported from the illuminated side to the shaded side of the shoot. (1)

The high auxin concentration on the shaded side promotes shoot growth. (1)

© 2019 Aristo Educational Press Ltd.

As a result, the shoot grows faster on the shaded side and it bends towards the light. (1)

(c) This brings the leaves to face the light source so that they can receive maximum light for photosynthesis. (1)

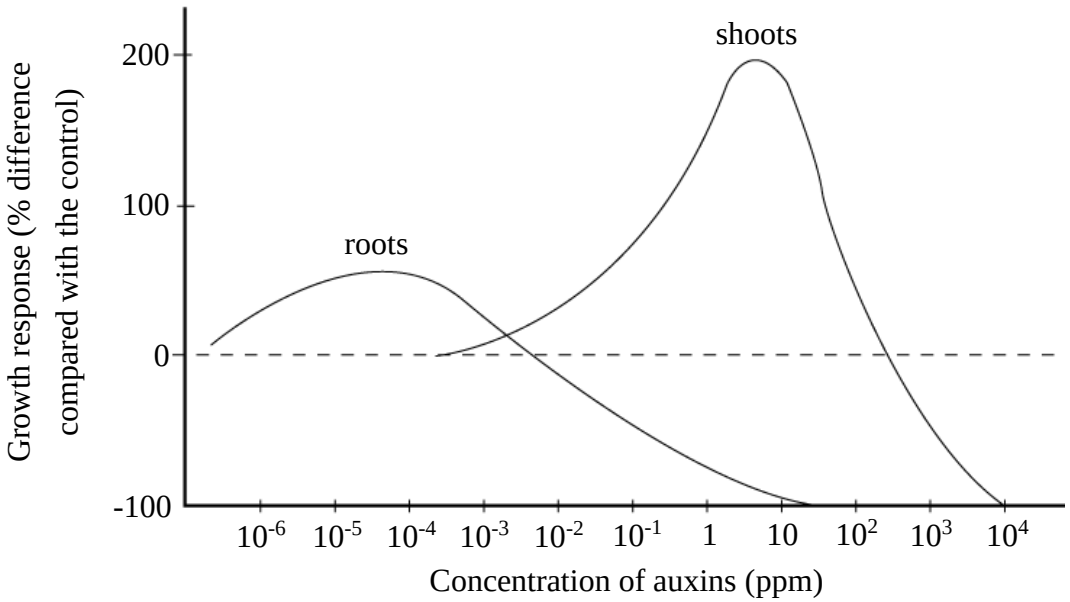
(5 marks)

##

	Skills assessed	Marks
(a)	Understanding basic concepts	1
(b)	Understanding basic concepts	3
(c)	Understanding basic concepts	1

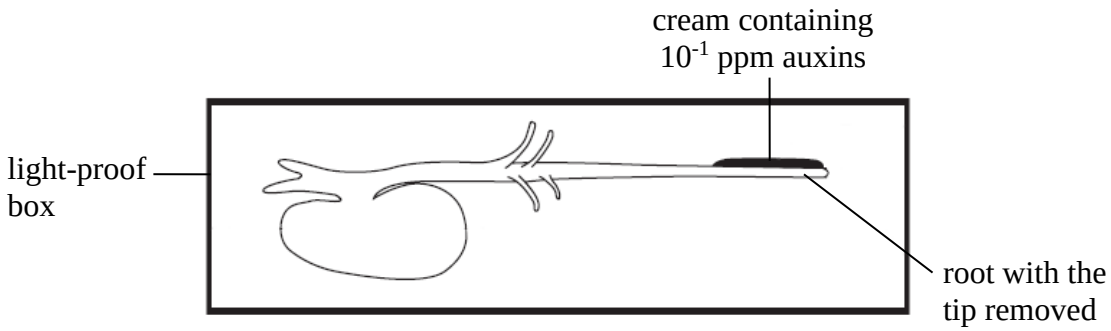
|!|BCESB0160000019|!| ★ **SI** **STSE**

The following graph shows the effects of auxin concentrations on the growth of shoots and roots of seedlings.



(a) Are the shoots or the roots more sensitive to auxins? (1 mark)

(b) A seedling, with its root tip removed, was placed horizontally in a light-proof box. A paste of cream containing 10^{-1} ppm auxins was applied on one side of the root as shown in the diagram below.



After two days, the root was found to bend upwards. Explain how this change in the root occurred with reference to the effect of auxins. (3 marks)

(c) Suggest ***one*** application of auxins in agriculture. (1 mark)

##

(a) Roots (1)

(b) Auxins at 10^{-1} ppm inhibit root growth. (1)

The upper side of the root received a higher auxin concentration than the lower side. (1)

The upper side of the root grew slower than the lower sider./ The lower side grew faster than the upper side. (1)

Thus, the root bent upwards.

(c) Auxins can be used in rooting powder to promote root growth.
(Or other reasonable answers) (1)

(5 marks)

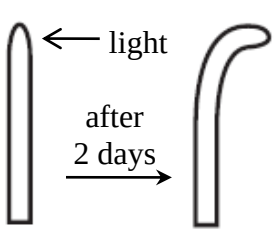
© 2019 Aristo Educational Press Ltd.

##

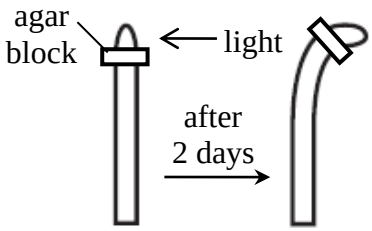
	Skills assessed	Marks
(a)	Analysing data	1
(b)	Analysing data	3
(c)	Applying concepts	1

|!|BCESB0160000020|!| ★ **SI**

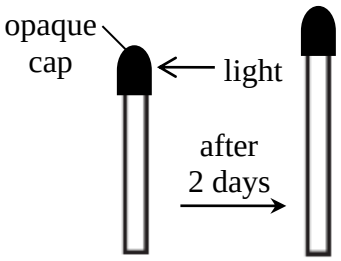
The diagram below shows the set-ups of an experiment and the results after two days



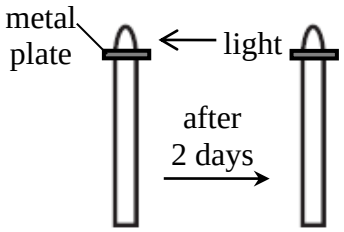
coleoptile 1



coleoptile 2



coleoptile 3



coleoptile 4

(a) From the results of the experiment, which part of the coleoptile is sensitive to unilateral light? Explain your answer. (2 marks)

(b) What conclusion can be drawn from the results of coleoptiles 2 and 4? (3 marks)

© 2019 Aristo Educational Press Ltd.

##

(a) The tip (1)

If the tip is covered by the opaque cap / blocked from the light, no bending growth movement occurs. (1)

(b) Some chemical substances produced from the tip of the coleoptiles (1)

can diffuse through the agar block (1)

to reach the lower part of the coleoptile and exert their effects there / cause the bending growth at the lower part of the coleoptile. (1)

(5 marks)

##

	Skills assessed	Marks
(a)	Analysing data	2
(b)	Analysing data	3

© 2019 Aristo Educational Press Ltd.